HARD Mock Set 2 — Accenture Technical (Above-Level)

Level: Harder than the 10 uploaded sets. No question here is repeated in Sets 1, 3, 4 or 5 — the whole 250-question bank is unique.

Format: 50 questions · Time: 50 minutes, closed book · No negative marking.

Section	Area	Qs
A	MS Office & Common Applications	8
B	Networking, Security & Cloud	12
C	CS Core — DSA / DBMS / OOP / HTML-CSS-JS	12
D	Pseudocode & Code Tracing	14
E	Applied — SQL & Code Output	4

Section A — MS Office & Common Applications (8)

Q1. A1:A6 = 10, 5, 20, 15, 30, 25 and B1:B6 = 1, 2, 3, 4, 5, 6. What does `=SUMIF(A1:A6, ">15", B1:B6)` return?

A) 21

B) 9

C) 6

D) 14

Q2. A1 contains `Amit Kumar Sharma`. What does `=MID(A1, FIND(" ", A1) + 1, 3)` return?

A) `"Kum"`

B) `"mit"`

C) `"Ami"`

D) `"Sha"`

Q3. What does `=ROUNDUP(-3.2, 0)` return in Excel?

A) –3

B) –3.2

C) –4

D) 0

Q4. You appended 50 new rows to the source data of a PivotTable, but it still shows the old totals. What must you do?

A) Press F9 — it recalculates everything

B) Nothing; PivotTables update automatically

C) Click **Refresh** (and extend the source range if the new rows fall outside it)

D) Delete and rebuild the PivotTable

Q5. `=IF(AND(A1>=50, B1<100), "OK", IF(OR(A1<20, B1>200), "ALERT", "REVIEW"))` with A1 = 60, B1 = 250. Output?

A) OK

B) REVIEW

C) #VALUE!

D) ALERT

Q6. You formatted chapter titles manually (bold, size 16) and inserted a Table of Contents — it comes out empty. Why?

A) The document is too long

B) The TOC field is broken

C) Word builds a TOC from **built-in Heading styles**, not manual formatting

D) You must enable Track Changes first

Q7. Which extension is a PowerPoint **template**?

A) `.pptx`

B) `.potx`

C) `.ppsx`

D) `.pptm`

Q8. Which Windows shortcut opens the **clipboard history** (multiple copied items)?

A) Win + V

B) Ctrl + V

C) Win + C

D) Ctrl + Shift + V

Section B — Networking, Security & Cloud (12)

Q9. What is the subnet mask for `/19`?

A) 255.255.192.0

B) 255.255.240.0

C) 255.255.224.0

D) 255.255.248.0

Q10. How many `/27` subnets can be carved out of one `/24`?

A) 4

B) 6

C) 8

D) 16

Q11. A branch LAN must support **60 hosts**. Which is the smallest (most efficient) prefix?

A) /25

B) /27

C) /28

D) /26

Q12. An engineer statically assigns `192.168.5.255/24` to a PC. What is wrong?

A) Nothing — it is a valid host address

B) It is the **broadcast address** of the subnet and cannot be assigned to a host

C) It is a Class B address

D) It is an APIPA address

Q13. On an 802.1Q trunk, frames belonging to the **native VLAN** are:

A) Dropped by default

B) Always tagged with VLAN ID 4095

C) Transmitted **untagged**

D) Tagged with VLAN ID 1 only if tagged explicitly

Q14. Three switches run STP: SW-A priority 32768, SW-B priority 32768 (MAC higher than SW-A), SW-C priority 4096. Which becomes the **root bridge**?

A) SW-A (lowest MAC)

B) SW-B

C) SW-C (lowest priority wins first)

D) Cannot be determined

Q15. A router learns 10.0.0.0/8 from static config, OSPF, EIGRP and RIP. Which route is installed?

A) OSPF (AD 110)

B) EIGRP (AD 90)

C) The static route (AD 1 — lowest administrative distance)

D) RIP (AD 120)

Q16. Correct order of the DHCP address-acquisition exchange:

A) Discover → Offer → Request → Acknowledge

B) Request → Offer → Discover → Acknowledge

C) Discover → Request → Offer → Acknowledge

D) Offer → Discover → Request → Acknowledge

Q17. Which DNS record type must be queried to find the **mail server** for a domain?

A) A

B) PTR

C) MX

D) CNAME

Q18. An attacker sends unsolicited ARP replies claiming the default gateway's IP maps to the attacker's MAC. This is:

A) DHCP starvation

B) ARP spoofing / poisoning (man-in-the-middle)

C) DNS amplification

D) SYN flood

Q19. "The vendor manages the runtime, middleware and operating system; you deploy and manage only your application and data." Which cloud model?

A) IaaS

B) SaaS

C) On-premises

D) PaaS

Q20. Which WPA3 feature protects against **offline dictionary attacks** even when a weak passphrase is used?

A) TKIP

B) SAE (Simultaneous Authentication of Equals)

C) WEP shared-key authentication

D) MAC filtering

Section C — CS Core: DSA / DBMS / OOP / HTML-CSS-JS (12)

Q21. The recurrence `T(n) = 2T(n/2) + O(n)` solves to:

A) O(n)

B) O(log n)

C) O(n log n)

D) $O(n^2)$

Q22. To maintain the **median of a continuously arriving stream** of numbers efficiently, you use:

A) A single min-heap

B) Two heaps — a max-heap for the lower half and a min-heap for the upper half

C) A hash table

D) A sorted array with binary search

Q23. Which statement about binary search is **FALSE**?

A) It requires the data to be sorted

B) It runs in O(log n) comparisons on an array

C) It needs random access to the middle element

D) It works efficiently on a **singly linked list** because it only needs sequential access

Q24. Keys 1, 2, 3, 4, 5, 6, 7 are inserted **in ascending order** into an initially empty BST. How many levels does the resulting tree have?

A) 3

B) 7

C) 6

D) 2

Q25. A relation has no partial or transitive dependencies but contains a non-trivial **multivalued dependency**. Which normal form is violated?

A) 3NF

B) 4NF

C) BCNF

D) 5NF

Q26. Using the table below, what is the **third row** returned by

`SELECT name, salary FROM Employees ORDER BY salary DESC, name ASC LIMIT 3;`?

emp_id	name	salary
1	Amit	50000
2	Bala	70000
3	Chitra	70000
4	Dev	45000
5	Esha	60000

A) Esha, 60000

B) Amit, 50000

C) Chitra, 70000

D) Dev, 45000

Q27. Which statement about database indexes is **FALSE**?

A) An index can speed up `WHERE` and `JOIN` lookups

B) An index on a column slows down `INSERT`/`UPDATE`/`DELETE` on that column

C) A clustered index determines the physical order of rows

D) Indexes improve every query and have **no** effect on write performance

Q28. Java:

```
String a = "Hi";
String b = "Hi";
String c = new String("Hi");
System.out.print((a == b) + " " + (a == c) + " " + a.equals(c));
```

A) `true true true`

B) `false false true`

C) `true false true`

D) `true false false`

Q29. Which HTML5 element should wrap a **self-contained** blog post that makes sense on its own?

A) `<section>`

B) `<article>`

C) `<div>`

D) `<aside>`

Q30. How many of these print `true`?

```
console.log([] == false);   // line 1
console.log(null == 0);     // line 2
console.log(NaN == NaN);    // line 3
```

A) 0

B) 2

C) 1

D) 3

Q31. What is `0x2F` in decimal?

A) 45

B) 57

C) 62

D) 47

Q32. Which HTTP method is **not** idempotent?

A) GET

B) PUT

C) DELETE

D) POST

Section D — Pseudocode & Code Tracing (14)

Q33.

```
count ← 0
FOR i ← 1 TO 6 STEP 2
    FOR j ← 1 TO i
        count ← count + 1
    END FOR
END FOR
PRINT count
```

A) 6

B) 12

C) 9

D) 21

Q34.

```
FUNCTION P(b, e)
    IF e = 0
        RETURN 1
    IF e MOD 2 = 0
        half ← P(b, e DIV 2)
        RETURN half * half
    RETURN b * P(b, e - 1)
END FUNCTION
PRINT P(2, 10)
```

A) 512

B) 100

C) 2048

D) 1024

Q35.

```
A ← [1, 2, 3, 4, 5]
n ← 5; k ← 2
FOR r ← 1 TO k
    temp ← A[0]
    FOR i ← 0 TO n - 2
        A[i] ← A[i + 1]
    END FOR
    A[n - 1] ← temp
END FOR
PRINT A[0], A[4]
```

A) 2 3

B) 3 5

C) 3 2

D) 5 1

Q36.

```
A ← [3, 1, 3, 2, 1, 3]
maxc ← 0; mode ← 0
FOR i ← 0 TO 5
    c ← 0
    FOR j ← 0 TO 5
        IF A[i] = A[j]
            c ← c + 1
        END IF
    END FOR
    IF c > maxc
        maxc ← c
        mode ← A[i]
    END IF
END FOR
PRINT mode, maxc
```

A) 1 3

B) 3 2

C) 2 1

D) 3 3

Q37.

```
i ← 10
DO
    i ← i + 2
WHILE i < 5
PRINT i
```

A) 10

B) 0

C) 12

D) Infinite loop

Q38.

```
s ← "  data  structures  "
count ← 0
prev ← ' '
FOR each ch IN s
    IF ch <> ' ' AND prev = ' '
        count ← count + 1
    END IF
    prev ← ch
END FOR
PRINT count
```

A) 3

B) 4

C) 1

D) 2

Q39.

```
n ← 29; i ← 2; flag ← TRUE
WHILE i * i <= n
    IF n MOD i = 0
        flag ← FALSE
        BREAK
    END IF
    i ← i + 1
END WHILE
PRINT flag, i
```

A) TRUE 5

B) TRUE 6

C) FALSE 29

D) TRUE 29

Q40. (Arrays are passed **by reference**)

```
FUNCTION bump(arr)
    FOR i ← 0 TO LENGTH(arr) - 1
        arr[i] ← arr[i] * 2
    END FOR
END FUNCTION
A ← [1, 2, 3]
bump(A)
PRINT A[0], A[2]
```

A) 1 3

B) 2 3

C) 2 6

D) Runtime error

Q41.

```
A ← [-2, 1, -3, 4, -1, 2, 1, -5, 4]
maxEnding ← A[0]; maxSoFar ← A[0]
FOR i ← 1 TO 8
    maxEnding ← MAX(A[i], maxEnding + A[i])
    maxSoFar ← MAX(maxSoFar, maxEnding)
END FOR
PRINT maxSoFar
```

A) 7

B) 4

C) 11

D) 6

Q42.

```
a ← 0; b ← 1
FOR i ← 1 TO 8
    t ← a + b
    a ← b
    b ← t
END FOR
PRINT a
```

A) 13

B) 21

C) 34

D) 55

Q43.

```
m ← 84; att ← 68
IF att < 75
    g ← "F"
ELSE IF m >= 90
    g ← "A"
ELSE IF m >= 80
    g ← "B"
ELSE
    g ← "C"
END IF
PRINT g
```

A) F

B) A

C) B

D) C

Q44.

```
n ← 29; c ← 0
WHILE n > 0
    n ← n AND (n - 1)
    c ← c + 1
END WHILE
PRINT c
```

A) 3

B) 5

C) 4

D) 29

Q45.

```
n ← 9875
WHILE n > 9
    s ← 0
    WHILE n > 0
        s ← s + n MOD 10
        n ← n DIV 10
    END WHILE
    n ← s
END WHILE
PRINT n
```

A) 9

B) 11

C) 29

D) 2

Q46.

```
count ← 0
FOR i ← 1 TO 3
    FOR j ← 1 TO 3
        IF j > i
            BREAK
        END IF
        count ← count + 1
    END FOR
END FOR
PRINT count
```

A) 9

B) 6

C) 3

D) 7

Section E — Applied: SQL & Code Output (4)

Tables for Q47–Q48:

`Departments(dept_id, dept_name)` → (1, HR), (2, IT), (3, FIN) `Employees(emp_id, name, dept_id)` → (1, Amit, 1), (2, Bala, 2), (3, Chitra, 2), (4, Dev, NULL)

Q47. How many rows does this return?

```
SELECT d.dept_name, e.name
FROM Departments d
INNER JOIN Employees e ON d.dept_id = e.dept_id;
```

A) 4

B) 2

C) 3

D) 5

Q48. What value does this aggregate query return?

```
SELECT COUNT(*) FROM Employees WHERE dept_id <> 2;
```

A) 2

B) 3

C) 4

D) 1

Q49. Python:

```
def add(x, lst=[]):
    lst.append(x)
    return lst

print(add(1), add(2))
```

A) `[1] [2]`

B) `[1] [1]`

C) `[1] [1, 2]`

D) `TypeError`

Q50. JavaScript:

```
let x = 5;
function f() { console.log(x); let x = 3; }
f();
```

A) 5

B) 3

C) `undefined`

D) `ReferenceError: Cannot access 'x' before initialization`

ANSWER KEY

Do not look here until you have finished all 50 questions.

— Answer Key for Set 2

Quick grid

Q	1	2	3	4	5	6	7	8	9	10
Ans	D	A	C	C	D	C	B	A	C	C

Q	11	12	13	14	15	16	17	18	19	20
Ans	D	B	C	C	C	A	C	B	D	B

Q	21	22	23	24	25	26	27	28	29	30
Ans	C	B	D	B	B	A	D	C	B	C

Q	31	32	33	34	35	36	37	38	39	40
Ans	D	D	C	D	C	D	C	D	B	C

Q	41	42	43	44	45	46	47	48	49	50
Ans	D	B	A	C	D	B	C	D	C	D

Explanations

Q1 — D. Rows where A > 15: 20 (B3 = 3), 30 (B5 = 5), 25 (B6 = 6) → 3 + 5 + 6 = **14**. SUMIF sums the *third* argument, not the tested range.

Q2 — A. `FIND(" ")` = 5 → `MID(A1, 6, 3)` = characters 6–8 of "Amit Kumar Sharma" = **"Kum"**.

Q3 — C. ROUNDUP rounds **away from zero**, so –3.2 → **–4** (not –3). `ROUND(-3.2,0)` would give –3.

Q4 — C. PivotTables do **not** auto-refresh; press Refresh (or Refresh All). F9 recalculates worksheet formulas, which is not the same thing. Also extend the source range / use a Table if new rows sit outside it.

Q5 — D. `AND(60>=50, 250<100)` → FALSE, so we fall to the second IF: `OR(60<20, 250>200)` → TRUE → **ALERT**.

Q6 — C. The TOC is generated from paragraphs using built-in Heading 1/2/3 styles. Manual bold + size 16 looks identical on screen but is invisible to the TOC field.

Q7 — B. `.potx` = template, `.pptx` = presentation, `.ppsx` = slide show, `.pptm` = macro-enabled presentation.

Q8 — A. Win + V opens clipboard history (once enabled); Ctrl + V is a normal paste of the last item only.

Q9 — C. /19 = 8 + 8 + 3 network bits → third octet 224 → **255.255.224.0**.

Q10 — C. /27 vs /24 = 3 extra network bits → 2^3 = **8** subnets.

Q11 — D. Need 2^h – 2 ≥ 60 → h = 6 (62 hosts) → prefix = 32 – 6 = **/26**.

Q12 — B. In 192.168.5.0/24 the last address (`.255`) is the **directed broadcast** address and is not assignable to a host.

Q13 — C. Native-VLAN frames cross an 802.1Q trunk **untagged**; a mismatch of native VLAN on the two ends causes VLAN hopping/leakage.

Q14 — C. Bridge ID = priority first, then MAC. SW-C (4096) beats the default 32768 regardless of MAC.

Q15 — C. Lowest administrative distance wins: static = 1 < EIGRP 90 < OSPF 110 < RIP 120.

Q16 — A. DORA: Discover → Offer → Request → Acknowledge. The client broadcasts Discover; the server offers; the client requests; the server acknowledges.

Q17 — C. `MX` = mail exchanger. `A`/`AAAA` = name→address, `PTR` = address→name (reverse), `CNAME` = alias→canonical name.

Q18 — B. ARP spoofing/poisoning — poisoning the ARP caches of hosts and the gateway to become a man-in-the-middle.

Q19 — D. PaaS (e.g. Azure App Service, Heroku). IaaS = you manage the OS too; SaaS = you only use the software.

Q20 — B. WPA3-Personal replaces the PSK 4-way handshake with **SAE** (dragonfly), so an offline dictionary attack on captured frames is not possible.

Q21 — C. Master theorem: a = 2, b = 2, f(n) = O(n) = Θ(n^{log_b a}) → **O(n log n)** (merge sort).

Q22 — B. Two heaps: max-heap for the lower half, min-heap for the upper half, rebalanced so their sizes differ by ≤ 1 → median in O(1), insert O(log n).

Q23 — D. Binary search on a singly linked list costs O(n) because reaching the middle needs traversal — no random access.

Q24 — B. Sorted insertion into an unbalanced BST degenerates into a chain of **7 levels** (height 6, search O(n)).

Q25 — B. Multivalued dependencies are removed by **4NF**. The relation can already be in BCNF and still violate 4NF.

Q26 — A. Order: Bala 70000, Chitra 70000 (tie broken by name ascending), then **Esha 60000** → third row = Esha.

Q27 — D. Indexes cost disk space and slow down writes (the index must be updated). They also do not help queries that must scan most rows.

Q28 — C. `a` and `b` are both string-literal interned references → `==` true. `new String(...)` forces a fresh object → `==` false, but `equals` compares content → true.

Q29 — B. `<article>` = self-contained, independently distributable content. `<section>` = a thematic grouping; `<aside>` = tangentially related.

Q30 — C. Only line 1: `[] == false` → `[]` coerces to `""` → `0`, `false` → `0` → **true**. `null == 0` → false (null only == null/undefined). `NaN == NaN` → always false.

Q31 — D. 0x2F = 2×16 + 15 = **47**.

Q32 — D. GET, PUT, DELETE, HEAD, OPTIONS are idempotent (repeating has the same effect). **POST** creates a new resource each time.

Q33 — C. i = 1, 3, 5 → inner runs 1 + 3 + 5 = **9**.

Q34 — D. Exponentiation by squaring: 2¹■ = **1024** (with only ~4 multiplications, not 10).

Q35 — C. Two left rotations: [1,2,3,4,5] → [2,3,4,5,1] → [3,4,5,1,2] → `A[0] = 3`, `A[4] = 2`.

Q36 — D. Counts: 3 → 3, 1 → 2, 2 → 1. i = 0 gives c = 3 > 0 → mode 3, maxc 3. i = 2 (value 3) gives c = 3, **not > 3**, so nothing changes (strict `>`). → `3 3`.

Q37 — C. A DO…WHILE always executes the body once, so `i` becomes 12 and only then is the condition tested → prints **12**.

Q38 — D. Count space→non-space transitions: "data" (1) and "structures" (2). Leading/trailing runs don't count because `prev` starts as a space and the trailing spaces never transition to non-space → **2**.

Q39 — B. i runs 2,3,4,5 (25 ≤ 29), then 6 makes `i*i = 36 > 29` and the loop exits with **i = 6**, flag still TRUE (29 is prime).

Q40 — C. Arrays are reference types: `bump` mutates the caller's array → [2,4,6] → `2 6`.

Q41 — D. Kadane's algorithm: best subarray 4, –1, 2, 1 → **6**.

Q42 — B. After 8 iterations: (a,b) = (1,1), (1,2), (2,3), (3,5), (5,8), (8,13), (13,21), (21,34) → `a = 21` (b = 34 = F■).

Q43 — A. The attendance test comes **first**, so 68 < 75 short-circuits to **F** even though 84 marks would be a B.

Q44 — C. Brian Kernighan: `n & (n-1)` clears the lowest set bit. 29 = `11101` has **4** set bits → 4 iterations.

Q45 — D. Digital root: 9+8+7+5 = 29 → 2+9 = 11 → 1+1 = **2**.

Q46 — B. Triangular count: i = 1 → 1, i = 2 → 2, i = 3 → 3 → **6**. The BREAK only skips the tail of each inner loop.

Q47 — C. Matches: HR–Amit, IT–Bala, IT–Chitra = **3**. Dev (NULL dept_id) never matches; FIN has no employees.

Q48 — D. `dept_id <> 2` is UNKNOWN for Dev's NULL → excluded. Only Amit (dept 1) qualifies → **1**. Use `WHERE dept_id <> 2 OR dept_id IS NULL` to keep NULLs.

Q49 — C. The default list is created **once** at function definition, so both calls share it → `[1] [1, 2]`. Use `lst=None` and create a new list inside.

Q50 — D. `let` is hoisted but stays in the **temporal dead zone** until initialisation, so the inner `x` shadows the outer one and logging before `let x = 3` throws a ReferenceError.

Score guide (Set 2)

Score	Verdict
43–50	Excellent — you are safely above Accenture level
34–42	Good — revise the missed topics and re-sit
25–33	Needs work — return to the notes, then retry this set after 3 days
< 25	Go back to chapters 01–06 before Set 3